

● HARD

● ADVANCED MATH

● ANSWER KEY

Nonlinear equations in one variable and systems of equations in two variables

06

6 answers · Rationale-rich · Teach from doc

Q1**A**

A. $\frac{1}{2}$

Choice A is correct. The given equation can be rewritten as $\frac{1}{(x+5)^2} = 4$. Multiplying both sides of this equation by $(x+5)^2$ yields $1 = 4(x+5)^2$. Dividing both sides of this equation by 4 yields $\frac{1}{4} = (x+5)^2$. Taking the square root of both sides of this equation yields $\frac{1}{2} = x+5$ or $-\frac{1}{2} = x+5$. Therefore, a possible value of $x+5$ is $\frac{1}{2}$.

Choices B, C, and D are incorrect and may result from computational or conceptual errors.

Q2**120**

The correct answer is 120. The solutions to a quadratic equation of the form $ax^2 + bx + c = 0$ can be calculated using the quadratic formula and are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 2$, $b = -8$, and $c = -7$. It follows that the solutions to the given equation are $x = \frac{8 \pm \sqrt{8^2 - 4(2)(-7)}}{2(2)}$, which is equivalent to $x = \frac{8 \pm \sqrt{64 + 56}}{4}$, or $x = \frac{8 \pm \sqrt{120}}{4}$. It's given that one solution to the equation $2x^2 - 8x - 7 = 0$ can be written as $\frac{8 - \sqrt{k}}{4}$. The solution $\frac{8 - \sqrt{120}}{4}$ is in this form. Therefore, the value of k is 120.

Q3**C****C.** II and III only

Choice C is correct. Subtracting the expression $x - 29$ from both sides of the given equation yields $0 = x - ax - 29 - x - 29$, which can be rewritten as $0 = x - ax - 29 + -1x - 29$. Since the two terms on the right-hand side of this equation have a common factor of $x - 29$, it can be rewritten as $0 = x - 29x - a + -1$, or $0 = x - 29x - a - 1$. Since $x - a - 1$ is equivalent to $x - a + 1$, the equation $0 = x - 29x - a - 1$ can be rewritten as $0 = x - 29x - a + 1$. By the zero product property, it follows that $x - 29 = 0$ or $x - a + 1 = 0$. Adding 29 to both sides of the equation $x - 29 = 0$ yields $x = 29$. Adding $a + 1$ to both sides of the equation $x - a + 1 = 0$ yields $x = a + 1$. Therefore, the two solutions to the given equation are 29 and $a + 1$. Thus, only $a + 1$ and 29, not a , are solutions to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**6**

The correct answer is 6. It's given that a line with equation $2y = 4.5$ intersects a parabola with equation $y = -4x^2 + bx$, where b is a positive constant, at exactly one point in the xy -plane. It follows that the system of equations consisting of $2y = 4.5$ and $y = -4x^2 + bx$ has exactly one solution. Dividing both sides of the equation of the line by 2 yields $y = 2.25$. Substituting 2.25 for y in the equation of the parabola yields $2.25 = -4x^2 + bx$. Adding $4x^2$ and subtracting bx from both sides of this equation yields $4x^2 - bx + 2.25 = 0$. A quadratic equation in the form of $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one solution when the discriminant, $b^2 - 4ac$, is equal to zero. Substituting 4 for a and 2.25 for c in the expression $b^2 - 4ac$ and setting this expression equal to 0 yields $b^2 - 4(4)(2.25) = 0$, or $b^2 - 36 = 0$. Adding 36 to each side of this equation yields $b^2 = 36$. Taking the square root of each side of this equation yields $b = \pm 6$. It's given that b is positive, so the value of b is 6.

Q5

C

c. $\frac{1}{2}$

Choice C is correct. Each fraction in the given equation can be expressed with the common denominator $x^2 - 9$. Multiplying $\frac{2x}{x+3}$ by $\frac{x-3}{x-3}$ yields $\frac{2x^2-6x}{x^2-9}$, and multiplying $\frac{1}{x-3}$ by $\frac{x+3}{x+3}$ yields $\frac{x+3}{x^2-9}$. Therefore, the given equation can be written as $\frac{4x^2}{x^2-9} - \frac{2x^2-6x}{x^2-9} = \frac{x+3}{x^2-9}$. Multiplying each fraction by the denominator results in the equation $4x^2 - (2x^2 - 6x) = x + 3$, or $2x^2 + 6x = x + 3$. This equation can be solved by setting a quadratic expression equal to 0, then solving for x. Subtracting $x + 3$ from both sides of this equation yields $2x^2 + 5x - 3 = 0$. The expression $2x^2 + 5x - 3$ can be factored, resulting in the equation $(2x - 1)(x + 3) = 0$. By the zero product property, $2x - 1 = 0$ or $x + 3 = 0$. To solve for x in $2x - 1 = 0$, 1 can be added to both sides of the equation, resulting in $2x = 1$. Dividing both sides of this equation by 2 results in $x = \frac{1}{2}$. Solving for x in $x + 3 = 0$ yields $x = -3$. However, this value of x would result in the second fraction of the original equation having a denominator of 0. Therefore, $x = -3$ is an extraneous solution. Thus, the only value of x that satisfies the given equation is $x = \frac{1}{2}$.

Choice A is incorrect and may result from solving $x + 3 = 0$ but not realizing that this solution is extraneous because it would result in a denominator of 0 in the second fraction. Choice B is incorrect and may result from a sign error when solving $2x - 1 = 0$ for x. Choice D is incorrect and may result from a calculation error.

Q6**D****D. 40**

Choice D is correct. Adding 10 to each side of the given equation yields $x^2 - 40x = 10$. To complete the square, adding $\frac{40^2}{2}$, or 20^2 , to each side of this equation yields $x^2 - 40x + 20^2 = 10 + 20^2$, or $x - 20^2 = 410$. Taking the square root of each side of this equation yields $x - 20 = \pm \sqrt{410}$. Adding 20 to each side of this equation yields $x = 20 \pm \sqrt{410}$. Therefore, the solutions to the given equation are $x = 20 + \sqrt{410}$ and $x = 20 - \sqrt{410}$. The sum of these solutions is $20 + \sqrt{410} + 20 - \sqrt{410}$, or 40.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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